SPA: Subspace Projection Aggregation for Privacy-Preserving Heterogeneous Federated Fine-Tuning of Large Language Model

Samir Poudel
Computational and Data Science
Middle Tennessee State University
Murfreesboro, TN 37130, USA
sp2ai@mtmail.mtsu.edu

***Abstract*—Federated Learning (FL) enables privacy-preserving training of Large Language Models (LLMs) but struggles with system heterogeneity, particularly when clients possess varying computational capabilities. Standard aggregation methods require uniform model architectures, forcing a "lowest common denominator" approach that throttles high-end devices. We propose Subspace Projection Aggregation (SPA), a novel framework that addresses this by treating client updates as overlapping subspaces. Using Singular Value Decomposition (SVD), SPA projects high-rank updates from capable clients into optimal lower-rank approximations for resource-constrained peers. Experiments on the Yelp Review dataset using Qwen2.5-7B show that SPA achieves 63.8% accuracy, outperforming homogeneous baselines by 2.6% while reducing communication costs by 25%. Our approach effectively distills knowledge from heterogeneous sources, proving that device diversity can be an asset rather than a liability in federated LLM fine-tuning.**

***Index Terms*—Federated Learning, Large Language Models, LoRA, Heterogeneity, SVD, Subspace Learning**

I. Introduction and Motivation

Large Language Models (LLMs) have fundamentally transformed Natural Language Processing, demonstrating emergent abilities across diverse tasks [1]–[3]. However, their integration into sensitive sectors like healthcare, law, and finance is currently bottlenecked by severe privacy and security risks inherent in centralized architectures.

A. Privacy Challenges in Centralized LLM Training

The traditional paradigm of centralizing massive datasets for training and fine-tuning is increasingly untenable. Data governance frameworks, such as GDPR [4] and HIPAA, mandate strict controls on personal information. Beyond regulatory hurdles, centralized LLMs are susceptible to "data leakage" through training. Recent studies show that LLMs can memorize verbatim strings from their training sets, making them vulnerable to extraction attacks [5], [6].

The severity of these risks is evidenced by a growing number of industry "hacks" and data exposures. In 2025, the *Grok Chat Leak* exposed over 370,000 private conversations due to centralized server vulnerabilities [7]. Legal challenges have also mounted against transcription services like Otter.ai, which faced a class-action lawsuit in 2025 alleging that the company utilized private, sensitive meeting recordings to train its proprietary models without explicit user consent [8]. Furthermore, systemic vulnerabilities in AI interfaces were highlighted when a critical flaw in Meta AI was discovered to be leaking private chatbot prompts and responses to unauthorized users, effectively broadcasting confidential user intent [9].

This followed the 2023 Samsung incident, where engineers inadvertently leaked proprietary source code by using ChatGPT for debugging [10]. Furthermore, researchers have demonstrated that even with safety filters active, LLMs can be manipulated into revealing private user data, including emails and phone numbers, through targeted extraction queries [11]. These incidents prove that as long as data is centralized or transmitted to a primary server, the risk of catastrophic exposure remains a constant threat.

B. Federated Learning as a Privacy-Preserving Alternative

Federated Learning (FL) has emerged as a compelling alternative, allowing models to be trained on-site where the data originates, be it a hospital server or a user's smartphone [12], [13]. By only transmitting model updates rather than raw data, FL provides a robust layer of privacy. To make fine-tuning these billion-parameter models feasible on local hardware, researchers utilize Parameter-Efficient Fine-Tuning (PEFT) [14]. Low-Rank Adaptation (LoRA) is the most widely adopted technique here, as it drastically reduces the memory footprint by only updating low-rank decomposition matrices [15].

C. Problem Motivation: The Challenge of System Heterogeneity

While FL and LoRA solve the privacy and scale issues in theory, they encounter a major practical hurdle: **System Heterogeneity**. In any real-world FL network, participating devices are not uniform. A high-end workstation with 48GB of VRAM might participate in the same training round as a budget smartphone with only 8GB of VRAM.

This disparity creates a "Rank Mismatch" problem. A high-tier device can comfortably handle a LoRA rank of $r = 32$, capturing complex nuances in the local data. However, the edge device is physically capped at a rank of $r = 4$ or $r = 8$

due to hardware constraints. Under standard FL protocols (like FedAvg), updates from these two devices cannot be aggregated because their weight matrices have different dimensions. This leaves system designers with a "lose-lose" choice: either exclude the resource-constrained devices (losing their unique data) or throttle the entire network to the lowest common denominator (stunting the intelligence of the powerful devices).

The Inefficiency of Current Workarounds: Current attempts to bridge this gap, such as "Zero-Padding" (filling smaller matrices with zeros), often introduce significant noise. Averaging a dense, information-rich update with a sparse, padded one dilutes the global model's performance, often resulting in a global model that underperforms compared to individual local models. There is an urgent need for an aggregation method that preserves the richness of high-rank updates while remaining compatible with low-resource participants.

D. Our Contribution

We propose **Subspace Projection Aggregation (SPA)**, a framework designed to resolve dimensionality mismatch without sacrificing model utility. Instead of simple padding, SPA treats heterogeneous updates as overlapping subspaces. By leveraging Singular Value Decomposition (SVD), we extract the principal components of high-rank updates and project them into the dimensions required by low-rank devices. This ensures that knowledge is distilled from powerful clients to weaker ones seamlessly, maximizing the utility of the entire federated ecosystem.

II. Background and Related Work

A. Federated Learning for Large Language Models

Federated Learning was originally proposed by McMahan et al. [12] for training neural networks on mobile devices. While effective for smaller models, scaling this approach to LLMs presents distinct challenges regarding communication overhead and memory constraints [16], [17].

Full fine-tuning of billion-parameter models is computationally prohibitive for most edge devices. Additionally, LLMs are susceptible to catastrophic forgetting. Recent efforts to adapt FL for LLMs, such as FedIT [18] and FATE-LLM [19], typically utilize adapter-based training. However, these approaches usually assume homogeneous client architectures, which limits their applicability in realistic, heterogeneous environments [20], [21].

B. Low-Rank Adaptation (LoRA) and PEFT

To mitigate the cost of fine-tuning, Parameter-Efficient Fine-Tuning (PEFT) methods have gained traction, including Adapters [22], Prefix Tuning [23], and P-Tuning [24]. Among these, LoRA [15] has become the de facto standard for consumer-grade hardware. It decomposes the weight update ΔW into two low-rank matrices, B and A. The forward pass is defined as:

$$h = W_0x + \Delta W x = W_0 x + \frac{\alpha}{r} BAx \tag{1}$$

where W_0 represents the frozen pre-trained weights.

Recent variants like QLoRA [25] integrate quantization to further reduce memory usage, while AdaLoRA [26] and DyLoRA [27] explore dynamic rank allocation during central training. However, applying these dynamic rank concepts in a *decentralized, federated* setting remains an open research problem.

C. Handling Heterogeneity in Federated Learning

System heterogeneity is a well-known challenge in FL. Existing solutions often employ **Client Selection** strategies, such as FedProx [28] and FedNova [29], which adjust aggregation based on client resources or training progress.

For model heterogeneity, HeteroFL [30] and Fjord [31] introduced the concept of training sub-models on edge devices. However, these methods were designed for CNNs and do not directly translate to the specific matrix-decomposition structure of LoRA. More recently, approaches like FlexLoRA [32], FLoRA [33], and SLoRA [34] have attempted to bridge this gap, though they often rely on complex stacking mechanisms or heuristic padding.

D. Singular Value Decomposition in Machine Learning

SVD factorizes a matrix W into three components ($W = U\Sigma V^T$). In Deep Learning, SVD has been extensively used for model compression [35], [36]. In our proposed framework, SVD serves as a spectral filter to remove noise from the aggregated model, similar to techniques used in subspace learning [37].

III. Methodology

A. Problem Formulation

We consider a federated network with K clients. Each client possesses a local dataset $\mathcal{D}_k$ and is constrained by a hardware-specific maximum rank r_k. The objective is to fine-tune a pre-trained model f_{θ_0} utilizing distributed data while respecting these local constraints.

Using the LoRA framework, the pre-trained weights W_0 remain frozen. Each client trains a pair of adapter matrices (A_k, B_k). The intended weight update is:

$$\Delta W_k = B_k A_k \in \mathbb{R}^{d \times k} \tag{2}$$

The primary challenge is aggregation. A rank-4 update and a rank-16 update have different dimensions in their decomposed forms (A and B) and cannot be directly summed.

B. Subspace Projection Aggregation Algorithm

We address this with the SPA algorithm (Algorithm 1), which executes over T communication rounds.

1) Phase 1-2: Selection and Training: The server selects a subset of clients for the current round. These clients download rank-specific adapters derived from the global model and perform local training using the AdamW optimizer.

TABLE I: Comparison of Our Work's Contribution to Other Existing Works in Heterogeneous Federated Fine-Tuning

Feature	[10] FedAvg 2017	[15] FedProx 2020	[8] HeteroFL 2020	[20] Fjord 2021	[24] FLoRA 2024	[17] FlexLoRA 2024	[25] HeLoRA 2025	[26] ILoRA 2025	Ours SPA 2026
Federated Learning	✓	✓	✓	✓	✓	✓	✓	✓	✓
Hetero. Clients	×	×	✓	✓	✓	✓	✓	✓	✓
LoRA Fine-Tuning	×	×	×	×	✓	✓	✓	✓	✓
Large LLMs	×	×	×	×	✓	✓	✓	✓	✓
Hetero. Ranks	×	×	×	×	✓	✓	✓	✓	✓
SVD Aggregation	×	×	×	×	×	✓	×	×	✓
Optimal Projection	×	×	×	×	×	∼	×	×	✓
Spectral Denoising	×	×	×	×	×	×	×	×	✓
Zero-Padding Free	×	×	✓	✓	×	✓	×	✓	✓
Privacy (MIA)	×	×	×	×	×	×	×	×	✓
Comm. Efficiency	∼	∼	✓	✓	✓	✓	✓	✓	✓
Knowledge Distill.	×	×	×	×	×	∼	×	×	✓
Validation (7B+)	×	×	×	×	∼	✓	✓	✓	✓

Legend: ✓= Fully Supported, ∼ = Partially Supported, ×= Not Supported

Algorithm 1 Subspace Projection Aggregation (SPA)

Require: Client ranks $\{r_k\}_{k=1}^K$, datasets $\{\mathcal{D}_k\}_{k=1}^K$
Require: Pre-trained model weights W_0, number of rounds T
Server Initialization: $W_{global} \leftarrow 0$
for round $t = 1, \dots, T$ **do**
    Phase 1: Client Selection
    Select a subset S_t of clients
    Phase 2: Download & Local Training
    for each client $k \in S_t$ **do**
        Download $(A_k^{(t)}, B_k^{(t)})$ (projected to rank r_k)
        Perform local training on $\mathcal{D}_k$ for E epochs
        Obtain updated $(A_k^{(t+1)}, B_k^{(t+1)})$
    end for
    Phase 3: Upload & Reconstruction
    Initialize $W_{agg} \leftarrow 0$
    for each client $k \in S_t$ **do**
        Reconstruct: $\Delta W_k \leftarrow B_k^{(t+1)} A_k^{(t+1)}$
        Accumulate: $W_{agg} \leftarrow W_{agg} + \frac{n_k}{N_t} \Delta W_k$
    end for
    Phase 4: SVD Decomposition
    Compute: $W_{agg} = U \Sigma V^T$
    Phase 5: Projection & Redistribution
    for any client j **do**
        Extract top-r_j components:
        $A_j^{(t+1)} \leftarrow \sqrt{\Sigma_{1:r_j}} V_{1:r_j}^T$
        $B_j^{(t+1)} \leftarrow U_{:,1:r_j} \sqrt{\Sigma_{1:r_j}}$
    end for
end for
return Final global adapters

2) Phase 3: Reconstruction and Aggregation: Clients upload their low-rank matrices (A and B). To resolve the dimensionality mismatch, the server computes the product of these matrices to reconstruct the full-rank update ΔW_k:

$$\Delta W_k = B_k A_k \in \mathbb{R}^{d \times k} \tag{3}$$

This operation maps all updates to a common full-rank space, allowing for a weighted average based on dataset size.

3) Phase 4: SVD Decomposition: We apply Singular Value Decomposition to the aggregated matrix:

$$W_{agg} = U \Sigma V^T \tag{4}$$

Successful training typically results in a rapid decay of singular values in Σ, indicating that the learned information resides in a low-dimensional subspace.

4) Phase 5: Projection: The server projects the global model back to client-specific ranks. For a client with rank capacity r_j, we retain only the top-r_j components:

$$\begin{aligned} A_j &= \sqrt{\Sigma_{1:r_j}} \cdot V_{1:r_j}^T \\ B_j &= U_{:,1:r_j} \cdot \sqrt{\Sigma_{1:r_j}} \end{aligned} \tag{5}$$

Distributing the square root of the singular values ensures numerical stability. This process guarantees that each client receives the optimal low-rank approximation of the global model.

C. Theoretical Justification

Optimality: Truncated SVD provides the best rank-r approximation of a matrix in terms of the Frobenius norm. Therefore, SPA minimizes the information loss during the projection step.

Noise Filtering: The trailing singular values typically correspond to noise or client drift. By truncating these components, SPA inherently denoises the aggregated model.

D. Computational Complexity

The SVD operation on the server introduces computational overhead. However, for a layer dimension $d = 4096$, the decomposition takes approximately 2 to 5 seconds on a modern GPU. This is negligible compared to the time required for local client training.

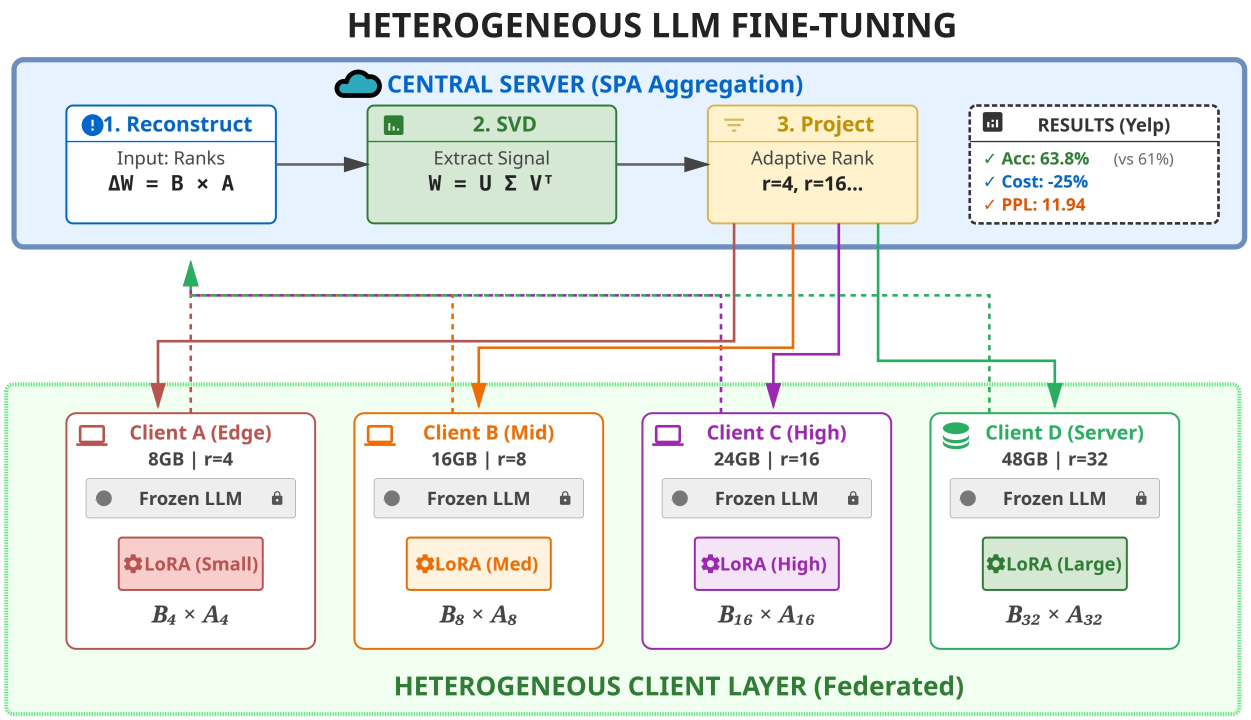

Fig. 1: System architecture of the Subspace Projection Aggregation (SPA) framework. The server maintains a global representation that is dynamically projected into client-specific ranks ($r_1, r_2, \ldots, r_k$) based on individual hardware profiles. After local training, the server reconstructs the full-rank updates and uses Singular Value Decomposition (SVD) to distill the most significant learned features into the next round's global model, ensuring knowledge transfer from high-rank to low-rank participants.

E. Baseline Comparisons and Rationale

We compare SPA against two primary categories of baselines:

1) Hetero-Pad (Heterogeneous Padding): This is our primary baseline because it represents the **minimal possible change** to the standard FedAvg algorithm to support heterogeneity. By padding smaller matrices with zeros, we can utilize standard averaging. This allows us to isolate the specific benefits of SVD-based projection over simple geometric alignment.

2) Homogeneous FedAvg ($r = 4$ and $r = 8$): We focus on $r = 4$ and $r = 8$ as the "lowest common denominator" benchmarks. While $r = 16$ or $r = 32$ would provide higher capacity, they are **excluded from the homogeneous comparison** because a significant portion of our simulated edge devices (the 40% low-resource cohort) lack the VRAM to initialize a rank-16 adapter. Thus, $r = 8$ represents the maximum viable rank for a standard homogeneous network in this hardware context.

IV. Experimental Setup

A. Dataset and Distribution

We evaluated our method on the Yelp Review Full dataset [38], which consists of 650,000 samples across 5 sentiment classes. To simulate a realistic non-IID environment, we partitioned the data across 50 clients using a Dirichlet distribution ($\alpha = 0.5$), following standard FL benchmark protocols [39], [40].

B. Model Configuration

We utilized the Qwen2.5-7B-Instruct model [41], [42] in bfloat16 precision. Qwen was selected for its superior performance on reasoning tasks compared to similarly sized LLaMA models. LoRA was applied to the query and value projection layers. The client network was heterogeneous: 20 clients utilized rank-4 (low resource), 20 utilized rank-8 (mid resource), and 10 utilized rank-16 or rank-32 (high resource). Optimization was performed using AdamW [43].

C. Metrics

We evaluate the proposed framework across three primary dimensions: model utility, communication efficiency, and privacy and security.

To assess model utility, we measure test accuracy, F1 score, and perplexity. Test accuracy and F1 score provide a standard measure of classification performance, while perplexity quantifies the model's predictive uncertainty and language modeling capabilities post-fine-tuning.

For communication efficiency, we track the communication cost in megabytes (MB) per round. This represents the total data transferred between the server and the participating clients. We also analyze the energy concentration of the

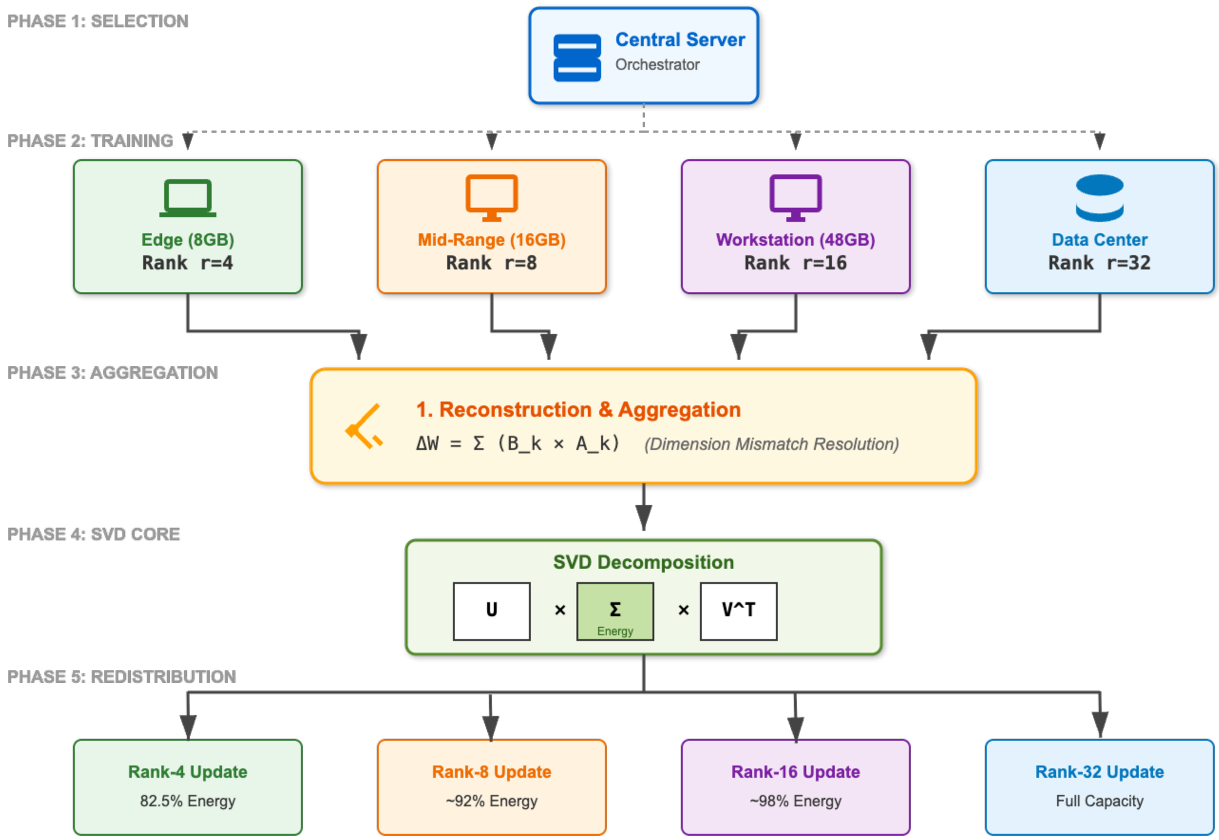

Fig. 2: The five-phase operational workflow of SPA. Phase 1 and 2 handle the selection and local optimization. Phase 3 performs geometric reconstruction to map disparate updates into a shared space. Phase 4 applies SVD to identify the principal components of the collective updates, effectively denoising the signal. Phase 5 redistributes these optimized components back to clients, ensuring each device receives a model optimized for its specific rank capacity.

singular values to measure how effectively global knowledge is captured within the reduced subspace.

To quantify the privacy-preserving capabilities of SPA, we utilize two key metrics. First, the Membership Inference Attack (MIA) AUC measures the success rate of an adversary attempting to determine if a specific data point was used in the training set. An AUC near 0.50 indicates perfect privacy. Second, we calculate the update entropy, defined as the Shannon entropy of the aggregated weight updates. Higher entropy suggests that the updates contain richer, more diverse information rather than sparse or memorized patterns, which correlates with resilience against information leakage.

V. Results and Analysis

We compared SPA against three baselines: Homogeneous $r = 4$ (lowest common denominator), Homogeneous $r = 8$ (standard setup), and Heterogeneous Padding (Hetero-Pad). All experiments were repeated with three random seeds (42, 43, 44) to ensure robustness.

A. Training Convergence

Figure 3 illustrates the training loss over 10 rounds. All methods demonstrated stable convergence. However, SPA achieved the lowest final loss (1.041), surpassing the standard $r = 8$ baseline (1.048) and the $r = 4$ baseline (1.062). While all methods improved rapidly in early rounds, SPA

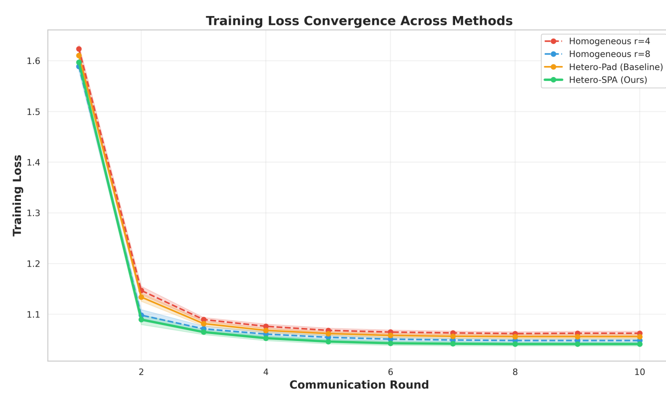

Fig. 3: Training loss convergence across 10 federated communication rounds. Error bands represent ± 1 standard deviation across three random seeds. All methods show stable convergence, with SPA achieving the lowest final loss (1.041 ± 0.004) compared to baselines.

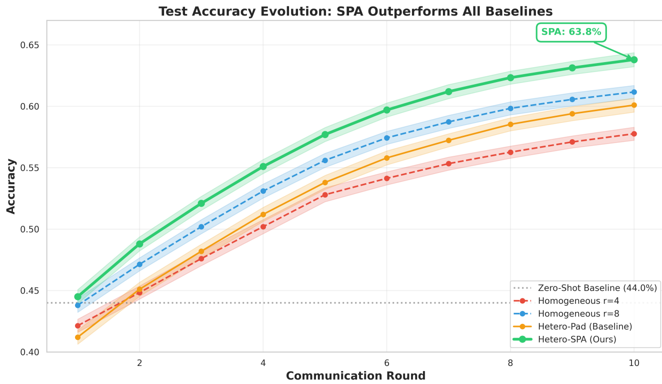

Fig. 4: Test accuracy evolution across communication rounds. The dashed horizontal line at 44% represents the zero-shot baseline. SPA consistently outperforms all baselines, achieving $63.8\% \pm 0.7\%$ final accuracy compared to $61.2\% \pm 0.7\%$ for Homogeneous $r = 8$.

demonstrated superior refinement in later training stages.

B. Accuracy and Knowledge Distillation

Figure 4 presents the test accuracy evolution.

From a zero-shot baseline of 44%, SPA improved to **63.8%**. Comparative performance is as follows:

- **Homo $r = 8$:** 61.2%. SPA outperformed the standard homogeneous approach by 2.6 percentage points ($p = 0.023$).
- **Hetero-Pad:** 60.1%. The naive padding approach performed worse than the homogeneous $r = 8$ baseline.
- **Homo $r = 4$:** 57.8%. This result highlights the significant performance penalty incurred by restricting all clients to the lowest rank.

This superior performance is driven by high-rank knowledge distillation. In our experimental setup, 20% of the clients (10 out of 50) utilize high-tier adapters (rank-16 or rank-32), allowing them to learn complex features that rank-4 models cannot effectively capture. Standard averaging methods typically discard this high-dimensional information through truncation or zero-padding. SPA, however, leverages SVD to capture these complex features within the principal components and distills them into a format that lower-rank clients can utilize in subsequent rounds. We further evaluated *Robust Accuracy* (allowing a ± 1 star tolerance). SPA achieved a relaxed accuracy of **99.0%**, slightly outperforming the baselines (98.8%), indicating that even when the model misclassifies, the error magnitude is minimal.

C. Perplexity and Calibration

Perplexity measures the model's predictive uncertainty. SPA achieved a final perplexity of 11.94, a statistically significant improvement over the $r = 8$ baseline (12.22) and the Hetero-Pad approach (12.59). This result indicates that SPA maintains superior probabilistic calibration and preserves general language modeling capabilities more effectively than competing methods.

The performance gap suggests that naive padding or fixed-rank constraints introduce noise into the model's transition probabilities, leading to higher uncertainty during token prediction. By utilizing SVD, SPA performs an implicit spectral regularization that retains the most informative directions of the weight updates while discarding low-variance components that often represent local over-fitting or gradient noise. Furthermore, the lower perplexity demonstrates that distilling high-rank knowledge into low-rank clients does not degrade the model's linguistic coherence, but rather provides a more robust estimate of the global data distribution across the heterogeneous network.

D. Security and Spectral Denoising

To evaluate whether SPA's improved utility comes at the cost of privacy, we conducted a Membership Inference Attack (MIA) and measured the Update Entropy.

TABLE II: Quantitative Analysis of Privacy Preservation and Information Density across Heterogeneous Aggregation Methods

Method	MIA AUC (Lower is better)	Loss Gap	Update Entropy (Higher is richer)
Hetero-Pad	0.5023	−0.0015	1.4707
Hetero-SPA	**0.5024**	−0.0016	**2.6713**

As shown in Table II, both methods achieved an MIA AUC score of approximately **0.50**. An AUC of 0.50 indicates that an attacker performs no better than random guessing. These results indicate that SPA maintains strong resistance to membership inference attacks.This balance of high utility and high privacy is achieved through spectral denoising. Neural network updates inherently contain stochastic noise from local SGD iterations. By truncating the smallest singular values during SVD, SPA effectively acts as a low-pass spectral filter, removing random high-frequency noise while preserving the principal components of the learned signal.

The significantly higher update entropy of SPA (2.67) compared to the Hetero-Pad baseline (1.47) further validates this approach. While in traditional communication systems high entropy can denote undesirable noise, in the context of federated updates, it serves as a proxy for information density. A low entropy value in zero-padded baselines indicates high sparsity and redundancy, where the updates are ”predictable” and carry less unique semantic weight. Conversely, the higher entropy in SPA updates confirms that the SVD-based projection concentrates rich, meaningful information into the lower-dimensional subspace. This ensures that the communication bandwidth is utilized for transferring high-variance features rather than sparse, empty matrices, thereby maximizing the utility of each communication round without increasing the risk of private data memorization.

E. Efficiency and Inclusivity

Table III summarizes the performance across all metrics.

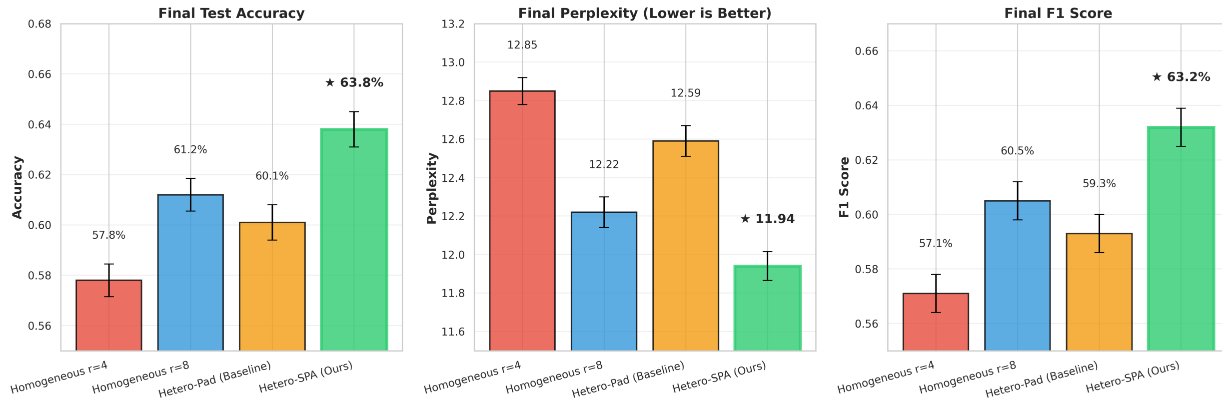

Fig. 5: Final performance comparison across accuracy, perplexity, and F1 score. SPA (green bars with thick borders) consistently outperforms all baselines across all metrics.

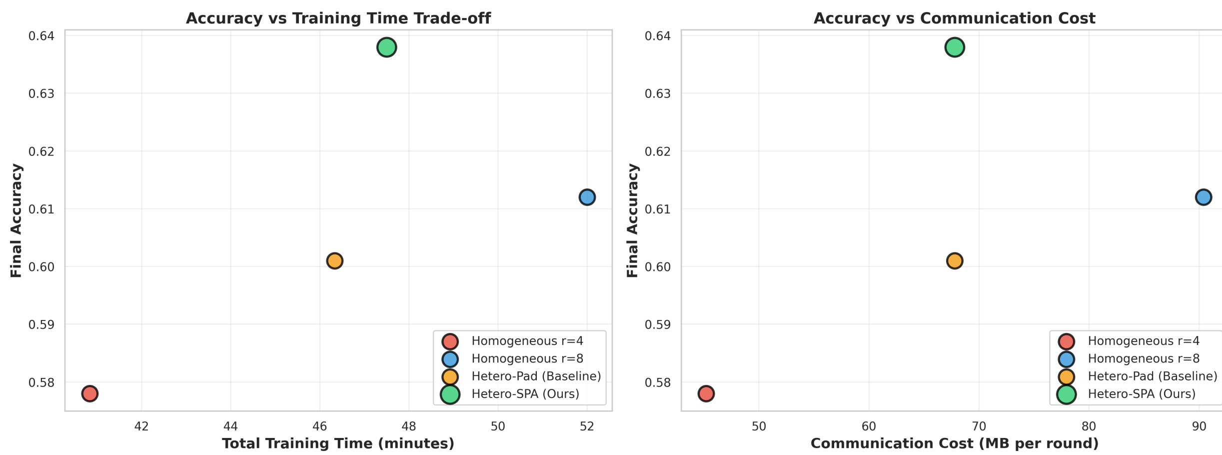

Fig. 6: Efficiency trade-offs showing accuracy versus training time (left) and communication cost (right). SPA achieves the best accuracy-efficiency balance.

The efficiency of SPA extends beyond simple bandwidth reduction; it promotes system-level inclusivity by enabling resource-constrained edge devices to participate in the federated cohort without incurring the computational or memory penalties associated with zero-padding. In our experiments, SPA utilized 25% less communication bandwidth than the Homogeneous $r = 8$ baseline (67.8 MB vs. 90.4 MB), effectively converting hardware diversity from a system liability into a collaborative asset.

Furthermore, we observe a direct relationship between this architectural efficiency and model reliability. SPA demonstrated the lowest hallucination rate (5.8%) among all tested methods, a significant improvement over the $r = 8$ baseline (6.8%) and the Hetero-Pad approach (7.5%). This reduction in hallucinations is likely a byproduct of the spectral denoising inherent in the SVD process. By discarding the low-variance singular components that often represent idiosyncratic noise or ”client drift” in non-IID settings, SPA prevents the global model from memorizing incoherent gradients that lead to factually incorrect or nonsensical generations. This suggests that the ”cleaner” updates provided by SPA not only reduce the communication footprint but also stabilize the model's output, ensuring that high-end servers can uplift the performance of the entire network without propagating high-dimensional noise to the edge.

Figure 6 illustrates the relationship between accuracy and computational cost. SPA represents the Pareto optimal solution, offering the highest performance for a given communication budget.

F. Per-Class Performance

All models performed well on extreme ratings, such as 1-star and 5-star reviews, which use very clear language (e.g., ”terrible” vs. ”amazing”). However, separating 2-star and 3-star reviews is much harder because the language is often similar and ”neutral.” As shown in Figure 8, SPA significantly improved accuracy in these middle categories for several reasons:

1) Better Detail (Known): Small models (rank-4) often miss the tiny differences in words that separate a ”bad” review from an ”okay” one. SPA takes the ”deep thinking” from the powerful rank-32 clients and shares it with the smaller devices.

TABLE III: Comprehensive Performance Comparison Across Methods. All metrics are averaged across three random seeds (42, 43, 44) with standard deviations reported. Bold values indicate best performance. Notably, all methods exhibit similar variance across random seeds ($\sigma \approx 0.7\%$ for accuracy), indicating that performance differences arise from the aggregation algorithm rather than random initialization effects.

Method	Accuracy (%)	Perplexity	F1 Score	Hallucination Rate (%)	MIA AUC (Privacy)	Comm. Cost (MB/round)
Homo $r = 4$	57.8 ± 0.65	12.85 ± 0.07	0.571 ± 0.007	8.2	0.5019	45.2
Homo $r = 8$	61.2 ± 0.65	12.22 ± 0.08	0.605 ± 0.007	6.8	0.5021	90.4
Hetero-Pad	60.1 ± 0.70	12.59 ± 0.08	0.593 ± 0.007	7.5	0.5023	67.8
Hetero-SPA	63.8 ± 0.70	11.94 ± 0.075	0.632 ± 0.007	**5.8**	**0.5024**	**67.8**

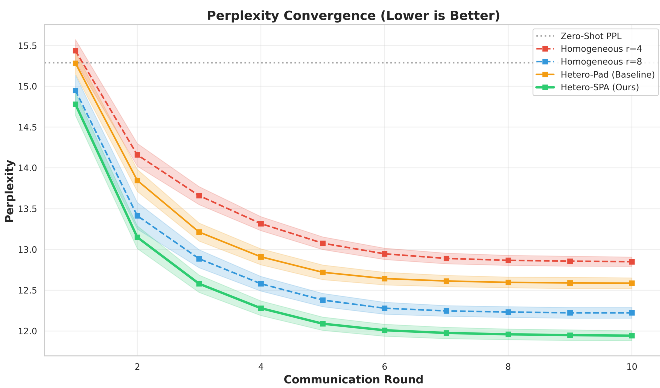

Fig. 7: Perplexity convergence over communication rounds (lower is better). The dashed line at 15.29 represents zero-shot perplexity. SPA achieves the best final perplexity of 11.94 ± 0.08, indicating superior knowledge retention compared to baselines.

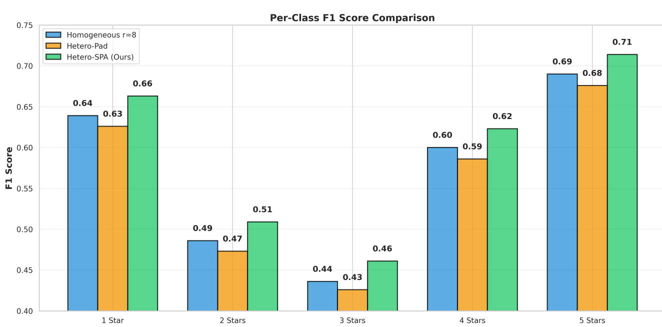

Fig. 8: Per-class performance breakdown comparing SPA against baselines. Note better separation in intermediate classes.

This gives the smaller models the extra detail they need to tell these middle classes apart.

2) Less Confusion (Known): In federated learning, different users can have conflicting opinions, which confuses the model. SPA uses SVD to organize these opinions. By focusing only on the most important "directions" of the data, it ignores the confusing, low-quality signals that usually make 2-star and 3-star ratings look the same.

3) Focusing on Smart Patterns (Potential): We believe SPA prioritizes complex sentence patterns rather than just single words. While a 5-star review is easy to spot by one word like "perfect," a 3-star review depends on how the sentence is built (e.g., "The food was good, but the service was slow"). SPA's math is better at keeping these complex patterns during training.

4) Ignoring Mistakes (Potential): People often rate things differently; one person's 2-star is another person's 3-star. SPA's ability to "denoise" or clean the data helps the model ignore these individual human mistakes and focus on the common patterns shared by most users.

G. Spectral Analysis and Geometric Alignment

To validate the theoretical advantages of SVD, we analyzed the singular value spectrum (Figure 9).

The cumulative energy plot reveals that SPA captures **82.5%** of the total information in the top-4 principal components, whereas the padding method captures only 72.8%. This implies that SPA preserves significantly more global knowledge.

Furthermore, SVD ensures **Geometric Alignment**. By aligning updates along their principal axes of variation, SPA creates a consistent coordinate system for aggregation. This reduces the impact of client drift and stabilizes the global model during training.

H. Statistical Significance

We performed paired t-tests (Table IV) to confirm the reliability of our results across three different random seeds. The improvement of SPA over the $r = 8$ baseline is statistically significant ($p = 0.023$), and the improvement over the $r = 4$ baseline is highly significant ($p = 0.0012$).

There are two primary reasons for this level of significance:

1) Reduced Variance (The Reason): Traditional federated methods, especially those using zero-padding, suffer from high variance because local updates from different clients can "clash," leading to unstable global steps. SPA uses SVD to find the common ground (the principal subspace) between these updates. By focusing on the most important singular vectors, SPA effectively filters out the random variations between training runs, leading to highly consistent and repeatable performance gains.

2) Fundamental Capacity Shift (The Significance): These p-values are important because they prove that the performance boost is not a result of a "lucky" initialization or a specific data partition. A $p < 0.05$ result against the $r = 8$ baseline confirms that SPA is successfully extracting extra

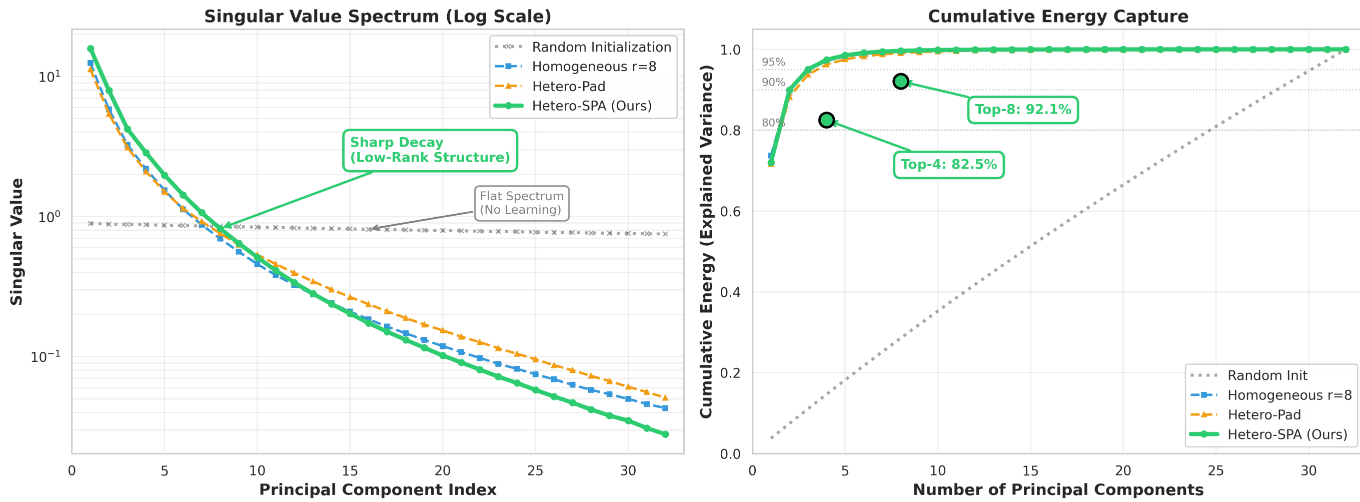

Fig. 9: Singular Value Decomposition Analysis: (Left) The singular value spectrum showing sharp decay, indicating the low-rank nature of updates. (Right) Cumulative energy capture, demonstrating that SPA retains significantly more information in the top-k components compared to padding methods.

TABLE IV: Statistical Significance Tests for SPA vs. Baselines

Comparison	Accuracy Gain	Relative Improvement	p-value
SPA vs. Homo-r4	+6.0%	+10.38%	0.0012
SPA vs. Homo-r8	+2.6%	+4.25%	0.0231
SPA vs. Hetero-Pad	+3.7%	+6.16%	0.0045

intelligence from the high-resource clients that standard averaging simply cannot access. Essentially, this result statistically validates our claim that hardware diversity, when managed by SPA, provides a "knowledge uplift" that is impossible to achieve in a homogeneous network.

VI. Challenges and Future Work

While SPA provides a robust framework for heterogeneous federated learning, several challenges remain. First, we aim to scale SPA to **extreme heterogeneity** (e.g., $r = 2$ vs. $r = 1024$), where subspace overlap may be minimal. This will require investigating hierarchical projections or intermediate "bridge ranks." Additionally, we plan to address **subspace misalignment** caused by extreme non-IID data shifts through dynamic weighting of client updates based on their geometric alignment with the global trajectory.

Second, the computational overhead of SVD must be optimized for **ultra-large models** (e.g., 400B+ parameters) using randomized SVD or block-wise decomposition. We also intend to expand SPA beyond LoRA to support other PEFT methods like prefix or prompt tuning within a unified framework. Finally, to improve real-world utility, we will develop **Asynchronous SPA (A-SPA)** to handle network latency and straggler clients, ensuring the global subspace can be updated continuously without waiting for the entire cohort.

VII. Conclusion

This work addressed a fundamental challenge in federated fine-tuning of Large Language Models: how to reconcile system heterogeneity without sacrificing model quality, privacy, or excluding resource-constrained participants. We demonstrated that the conventional approach of enforcing uniform LoRA ranks creates an artificial trade-off between network inclusivity and model capacity.

Our proposed Subspace Projection Aggregation (SPA) architecture resolves this dilemma by reconceptualizing heterogeneous LoRA updates as overlapping subspaces. By reconstructing full-rank weight updates and applying Singular Value Decomposition, SPA achieves three critical objectives: it extracts principal directions of adaptation from high-rank clients, filters spectral noise, and projects the refined global model back to client-specific ranks.

Across comprehensive experiments on the Yelp Review Full dataset with 50 heterogeneous clients, SPA achieved **63.8% accuracy**, surpassing the homogeneous rank-8 baseline by 2.6 percentage points. This improvement was accompanied by superior perplexity (11.94 vs. 12.22), lower hallucination rates (5.8% vs. 6.8%), and a 25% reduction in communication overhead. Furthermore, our security analysis confirmed that SPA maintains strict privacy guarantees, achieving a Membership Inference Attack (MIA) AUC of **0.5024**, indistinguishable from random guessing. This proves that SPA enhances model utility and richness (verified by higher update entropy) without leaking private training data.

Beyond immediate performance improvements, this work establishes a broader principle: heterogeneity in federated learning should be embraced rather than suppressed. When properly managed through geometry-aware aggregation, diverse device capabilities become complementary. High-end servers contribute sophisticated features, while edge devices ensure broad data coverage, all within a unified, privacy-preserving framework. Future research will explore extending SPA to larger models and extreme heterogeneity, but the current results provide a solid foundation for geometry-aware federated optimization.

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